

Tezpur University
Spring Semester End Examination-2025

Course Code: CS625

Course Title: Web Technology

Full marks: 30

Time: 90 mins

(The figures in the right-hand margin indicate full marks for the questions)

Q 1. Choose the correct option for the below multiple choice questions.

5X1=5 Marks

- (a) Proxy server is used to transform _____ protocol to _____ format.
(TCP-IP, OSI / OSI, TCP/IP / Non-HTTP, HTTP TCP/IP, HTTP)
- (b) In a client-server interaction, the _____ monitors the state of clients for any possible changes to take appropriate actions.
(HTTP response from the server / Server push method / Web Server / Object Request Broker)
- (c) A browser running an Active Web page is _____.
(a Thin-Client / a Active-Client / a Passive-Client / a Fat-Client)
- (d) Hidden variables travel with _____.
(no HTTP request/response pairs / some HTTP request/response pairs / only with HTTP requests / all HTTP request/ response pairs)
- (e) Non-repudiation of a message is best guaranteed only by _____.
(Certification Authority / Symmetric cryptography / Digital Certificate / Asymmetric cryptography)

Q 2. State True or False

5X1=5 Marks

- (a) TELNET uses IAC as the special control character to distinguish between control characters and data characters.
- (b) Only adding clients in a C/S system is an example of Horizontal Scaling.
- (c) A database server is accessed first before a transaction server to serve for any query request from a client.
- (d) The problem with dynamic URLs is the sensitivity of information.
- (e) When only the sender and the receiver want to be able to access the contents of a message, the principle of authorization comes into picture.

Q 3. Construct Render tree for the below HTML and CSS code. You also need to show the DOM tree and CSSOM tree during the construction of the Render tree.

5 Marks

HTML code

```
<html lang="en">
<head>
  <title>Web Technology</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>
    Web Technology
    <br> End Term Examination
    <br> Spring Semester 2025
  </h1>
  <div> </div>
  <p> CO423
    <span>CS625</span>
  </p>
</body>
</html>
```

CSS code

```
body { font-size: 16px;
}

img { width: 100px;
      float: right;
}

span { display: none;
       font-size: 36px;
       font-weight: bold;
}

h1 { font-size: 20px; color: blue; }

p { font-size: 20px; color: blue; }
```

- Q 4. (a) What are the advantages of Dynamic Web pages?
 (b) Discuss with the help of a scenario how a Web client can be made dynamic. Illustrate using figure and scripts.
2 Marks
- Q 5. (a) What are session variables?
 (b) Distinguish between cookies and session variables?
3 Marks
1 Mark
4 Marks
- Q 6. "The problem of lack of continuity from an application's point of view stills exists despite having persistent TCP connections."
 Analyze with proper justifications to uphold the truth of the above statement. Include a scenario for the analysis and the solution to it.
5 Marks

OR

- "Digital certificate is used for authenticating a Web client or Web server."
 Analyze with proper justifications to uphold the truth of the above statement. Show in the analysis how digital certificate helps in the secure generation of the secret key in a SSL handshake mechanism.
5 Marks

***** END OF PAPER *****

Quick CO Mapping

Q No.	Q1	Q2	Q3	Q4		Q5		Q6
				(a)	(b)	(a)	(b)	
-	CO1	CO1	CO2	CO1	CO1	CO1	CO3	CO3

Course Outcomes:

- CO1 (Understand) : Understand key Internet technologies supporting Internet applications.
 CO2 (Apply) : Implement web-based applications using suitable client-side and server-side web technologies such as HTML, CSS, XML, JavaScript, JSP, PHP, etc.
 CO3 (Analyze) : Differentiate between various methods, tools, frameworks, and web services used in web technologies.